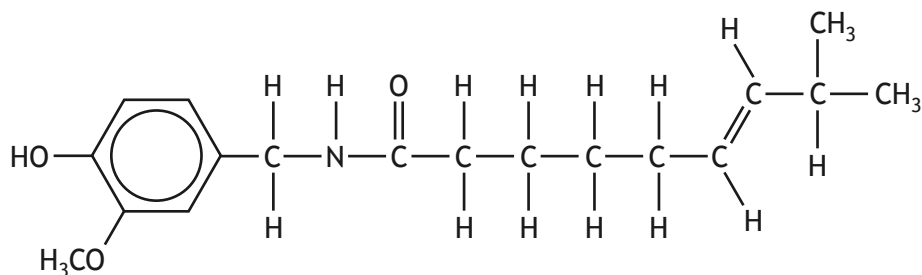
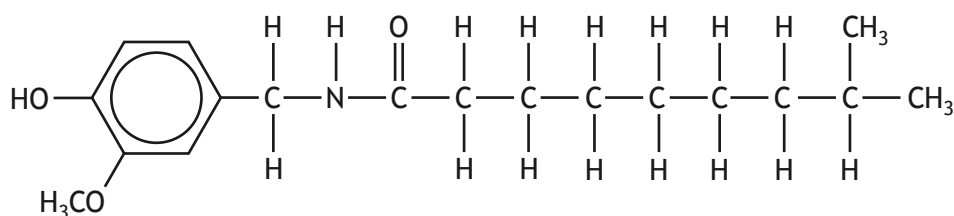


10. Chill peppers produce a burning sensation in the mouth. Chemists have identified a group of compounds found in chilli peppers called capsaicinoids that cause this sensation.

(a) The structures of two capsaicinoids are shown below.



capsaicin (CAP)



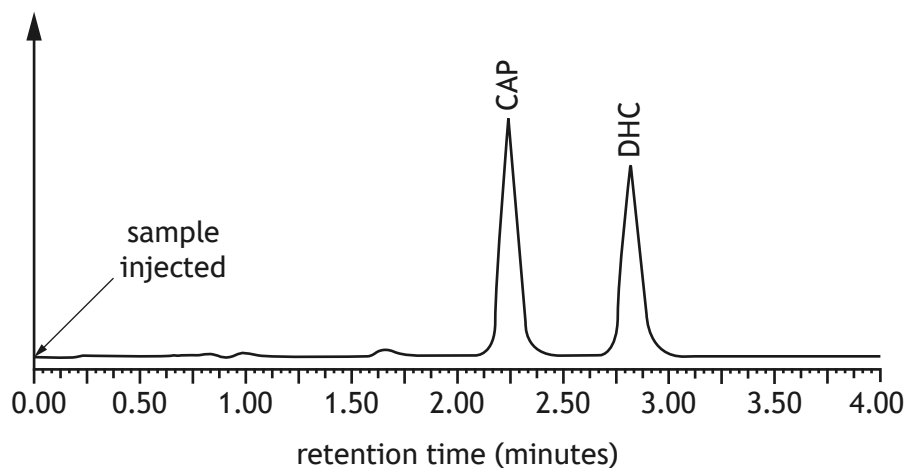
dihydrocapsaicin (DHC)

- (i) Circle the amide link in the CAP structure above.
(An additional structure, if required, can be found on page 35.)
- (ii) Describe a chemical test, and the expected result, that could be used to distinguish between CAP and DHC.
- (b) Taking a drink of water has little effect on the burning sensation in the mouth produced by capsaicinoids.
Suggest why the burning sensation produced by capsaicinoids is not affected by drinking water.

10. (continued)

- (c) High pressure liquid chromatography, HPLC, can be used to separate different capsaicinoids present in chilli peppers.

The time taken for a substance to travel through the HPLC apparatus is known as its retention time and is the time between the sample being injected and the centre of the peak.



Determine the retention time, **in seconds**, of CAP.

1

[Turn over



10. (continued)

- (d) Creams containing CAP are sold for pain relief.

A 60 g tube of cream contains 0.025% CAP.

CAP costs £1930.32 for 1.0 kg.

Calculate the cost, in pence, of the CAP in one tube of cream.

2

- (e) Chilli peppers are a good source of vitamin C, $C_6H_8O_6$.

Complete the ion electron equation for the oxidation of vitamin C.

1



* X 8 1 3 7 6 0 1 3 0 *